

CampusMate: A Prolog-Based Expert System for New Student Assistance

Prepared by: Data Mind Team

Date: October 15, 2025

1. Project Overview

When new students join a campus, they face many challenges such as finding buildings, understanding procedures, or accessing important contacts. This project aims to develop an Expert System using Prolog that provides instant answers to common campus-related questions.

The system will store structured knowledge about campus locations, services, and procedures in the form of facts and rules. Using Prolog's inference engine, the system will give intelligent responses to user queries through a simple text interface.

2. Problem Statement

- New students often find it difficult to get accurate campus information quickly.
- Human help desks are not available all the time and can be crowded.
- There is no intelligent system to provide 24/7 guidance.
- Manual searching wastes time and causes confusion.

3. Objectives

- To build a Prolog-based expert system to support new students.
- To provide information about campus locations, services, and emergency contacts.
- To create an interactive, easy-to-use command-line interface.
- To reduce dependency on administrative staff for basic queries.

4. Scope of the Project

- Campus buildings and their locations (library, labs, hostel, etc.)
- Service centers (registration, ID card, canteen, clubs)
- Emergency contacts and useful phone numbers
- Common questions and automated answers
- Simple interaction through terminal using Prolog queries.

5. Proposed Methodology

1. Requirement Gathering – Collect data about campus, facilities, and services.
2. Knowledge Base Design – Write facts and rules in Prolog.

3. System Development – Build interaction using Prolog queries and outputs.

4. Testing – Verify system accuracy with sample queries.

5. Deployment – Make it available for students on campus terminals or computers.

6. Tools & Technologies

- Programming Language: Prolog
- Interpreter: SWI-Prolog
- Platform: Command Line Interface (CLI)
- Editor: VS Code / SWI-Prolog IDE

7. Expected Outcome

- A functional expert system that gives correct answers to student questions.
- Reduced waiting time for basic information.
- Better campus navigation and student experience.
- Easy system expansion by adding more facts and rules.

8. Project Timeline (Estimated)

Phase: Requirement Gathering — **Duration:** 1 week — Collect campus data and FAQs

Phase: Knowledge Base Design — **Duration:** 2 weeks — Write facts and rules in Prolog

Phase: Development — **Duration:** 2 weeks — Build Prolog interface and reasoning engine

Phase: Testing — **Duration:** 1 week — Test with various queries

Phase: Deployment & Documentation — **Duration:** 1 week — Finalize and prepare for use

9. Future Enhancements

- Add voice or chatbot interface.
- Include live event and timetable information.
- Integrate multiple language support.
- Connect with web or mobile platforms in future.

10. Conclusion

This project will help new students quickly access important campus information through a Prolog expert system. It will reduce confusion, save time, and create a smarter and more welcoming campus environment.

11. References

- Prolog Programming for Artificial Intelligence.
- SWI-Prolog documentation.
- Campus handbooks and administrative documents.

Appendix: 4-Hour Hackathon Plan

Project: CampusMate — Data Mind Team

Date: October 15, 2025

Total Duration: 4 Hours

Hour 1: Requirement Setup & Knowledge Gathering (60 mins)

- Team Briefing – 10 mins (assign roles: knowledge engineer, developer, tester, documenter)
- Scope Finalization – 10 mins
- Data Collection – 30 mins (essential campus info)
- Prolog Environment Setup – 10 mins

Hour 2: Knowledge Base Design (60 mins)

- Write campus facts (library, hostel, canteen, contacts)
- Define rules (location_of/2, service_info/2, etc.)
- Test individual facts/rules in SWI-Prolog

Hour 3: System Development (60 mins)

- Develop user query interface (start/1, process/1 predicates)
- Integrate knowledge base and interface
- Basic error handling for unknown inputs

Hour 4: Testing, Debugging & Presentation (60 mins)

- Functional testing with sample queries
- Fix syntax and rule logic issues
- Prepare a 2–3 slide demo script (Problem → Solution → Demo → Future Enhancements)
- Final demo run